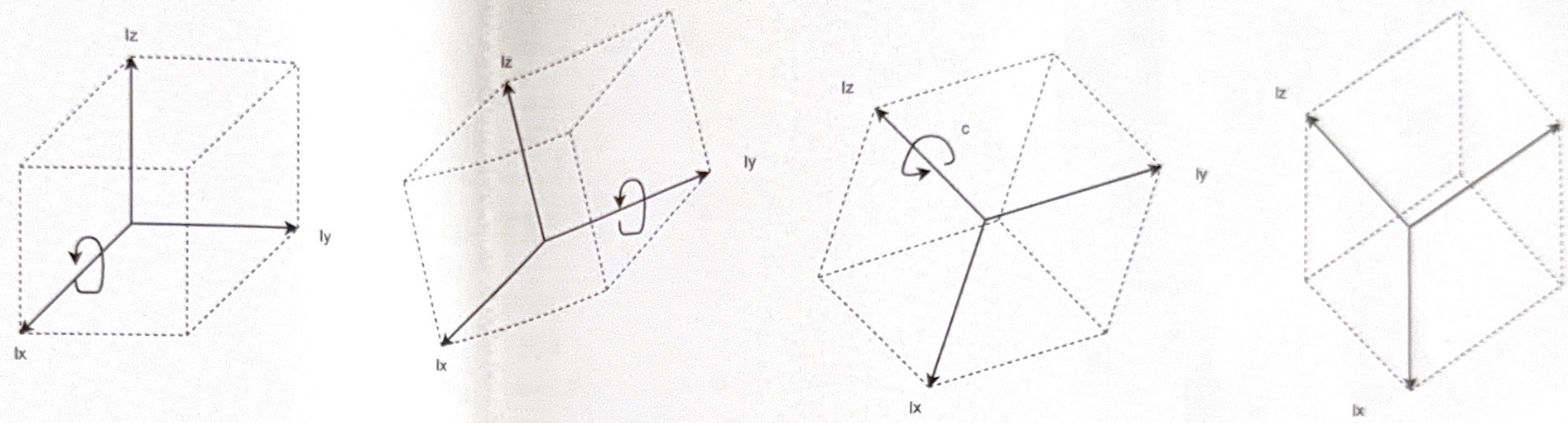
ISO 8855 coordinate system



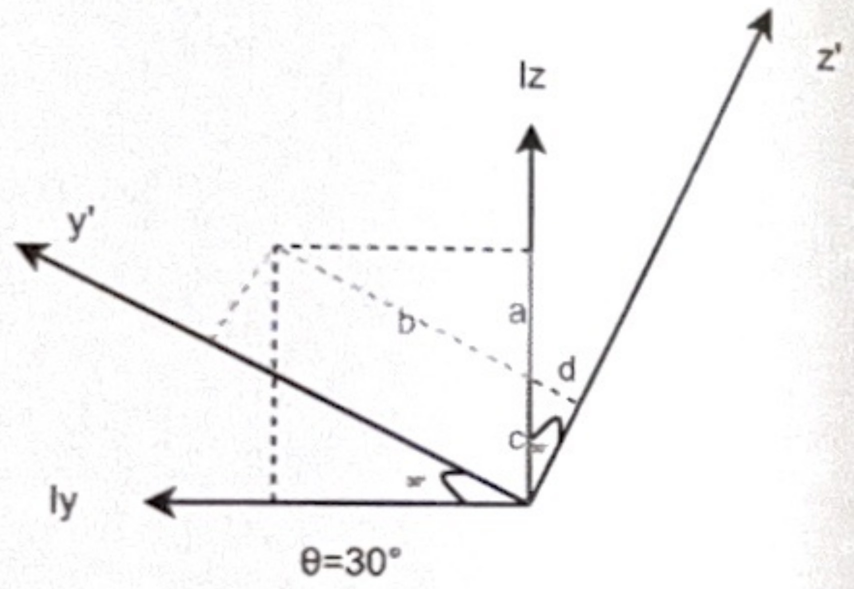
将Ri坐标按如下顺次旋转至R'
注意：所有旋转角为沿着轴的正方向作顺时针旋转

绕Vx旋转30°
绕Vy旋转60°
绕Vz旋转45°

以"绕某一轴做平面旋转时，点在该轴坐标不变"为原则进行手算

以单位坐标p0=(1,1,1)为例
最终计算得出旋转后坐标P1=(1.096,0.837,1.049)

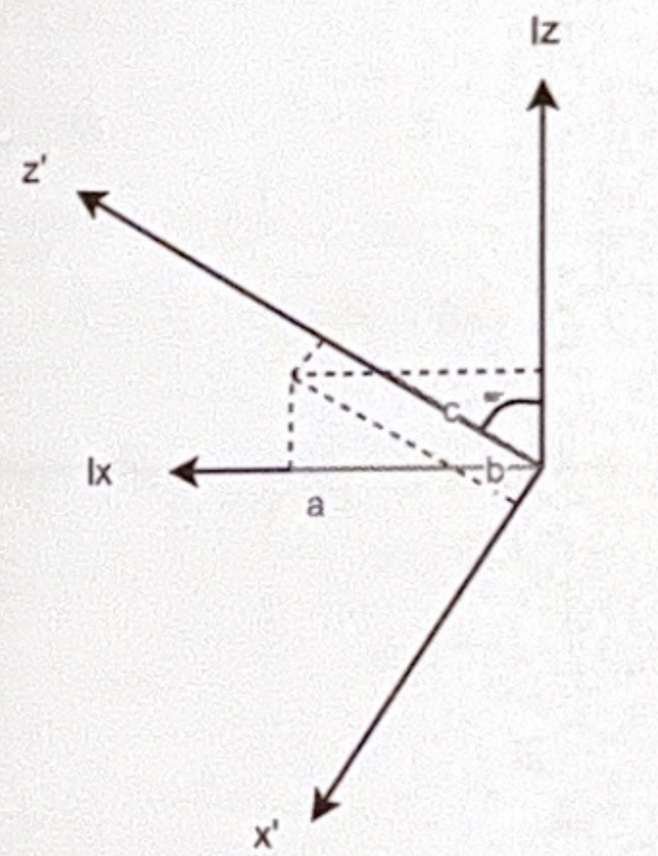
即对于加速度Rv=RT*RI，角速度同理



θ=30°

a=tanθ
b=1/cosθ
c=1-a
d=c*sinθ
y'=b+d
z'=c*cosθ

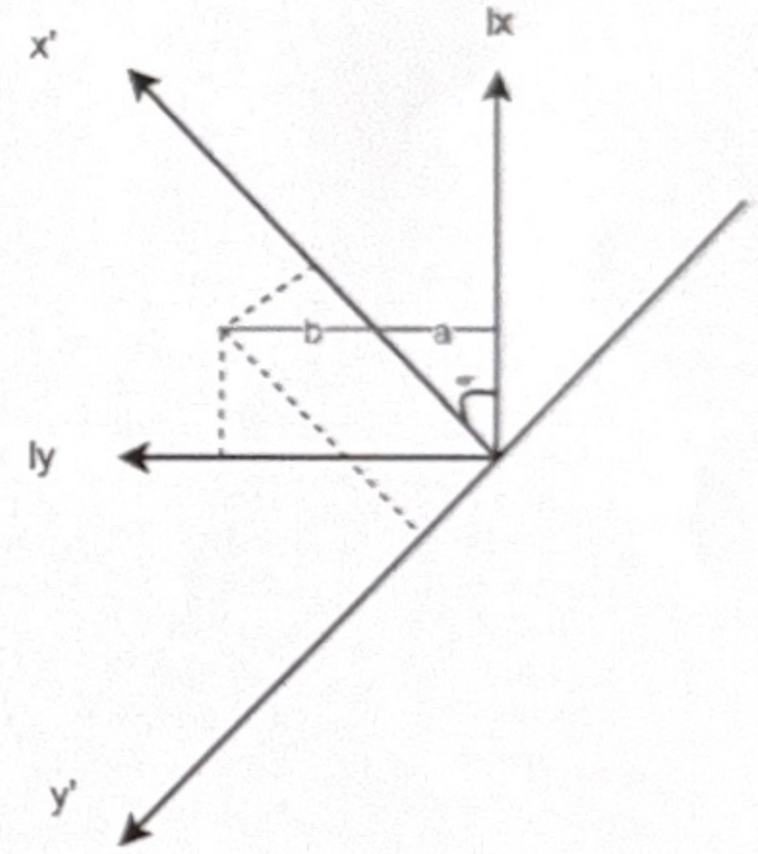
(1,1,1) -> (1,1.366,0.366)



θ=60°

a=Iz*tanθ
b=1-a
c=0.366/cosθ
x'=b*cosθ
z'=c+b*sinθ

(1,1.366,0.366) -> (0.183,1.366,1.049)



θ=45°

a=Ix
b=Iy-a
y'=b*sinθ
x'=a/sinθ + y'

(0.183,1.366,1.049) -> (1.096,0.837,1.049)

$$P0 := \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\theta_x := \frac{pi}{6}$$

$$\theta_y := \frac{pi}{3}$$

$$\theta_z := \frac{pi}{4}$$

$$R_x := \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta_x) & -\sin(\theta_x) \\ 0 & \sin(\theta_x) & \cos(\theta_x) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{pmatrix}$$

$$R_y := \begin{pmatrix} \cos(\theta_y) & 0 & \sin(\theta_y) \\ 0 & 1 & 0 \\ -\sin(\theta_y) & 0 & \cos(\theta_y) \end{pmatrix} = \begin{pmatrix} 0.5 & 0 & 0.866 \\ 0 & 1 & 0 \\ -0.866 & 0 & 0.5 \end{pmatrix}$$

$$R_z := \begin{pmatrix} \cos(\theta_z) & -\sin(\theta_z) & 0 \\ \sin(\theta_z) & \cos(\theta_z) & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.707 & -0.707 & 0 \\ 0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

P0依次绕XYZ轴旋转到P1坐标系

$$Rxyz := R_x \cdot R_y \cdot R_z = \begin{pmatrix} 0.354 & -0.354 & 0.866 \\ 0.919 & 0.306 & -0.25 \\ -0.177 & 0.884 & 0.433 \end{pmatrix}$$

$$Rxyz^{-1} = \begin{pmatrix} 0.354 & 0.919 & -0.177 \\ -0.354 & 0.306 & 0.884 \\ 0.866 & -0.25 & 0.433 \end{pmatrix}$$

$$P1 := Rxyz^{-1} \cdot P0 = \begin{pmatrix} 1.095 \\ 0.837 \\ 1.049 \end{pmatrix}$$

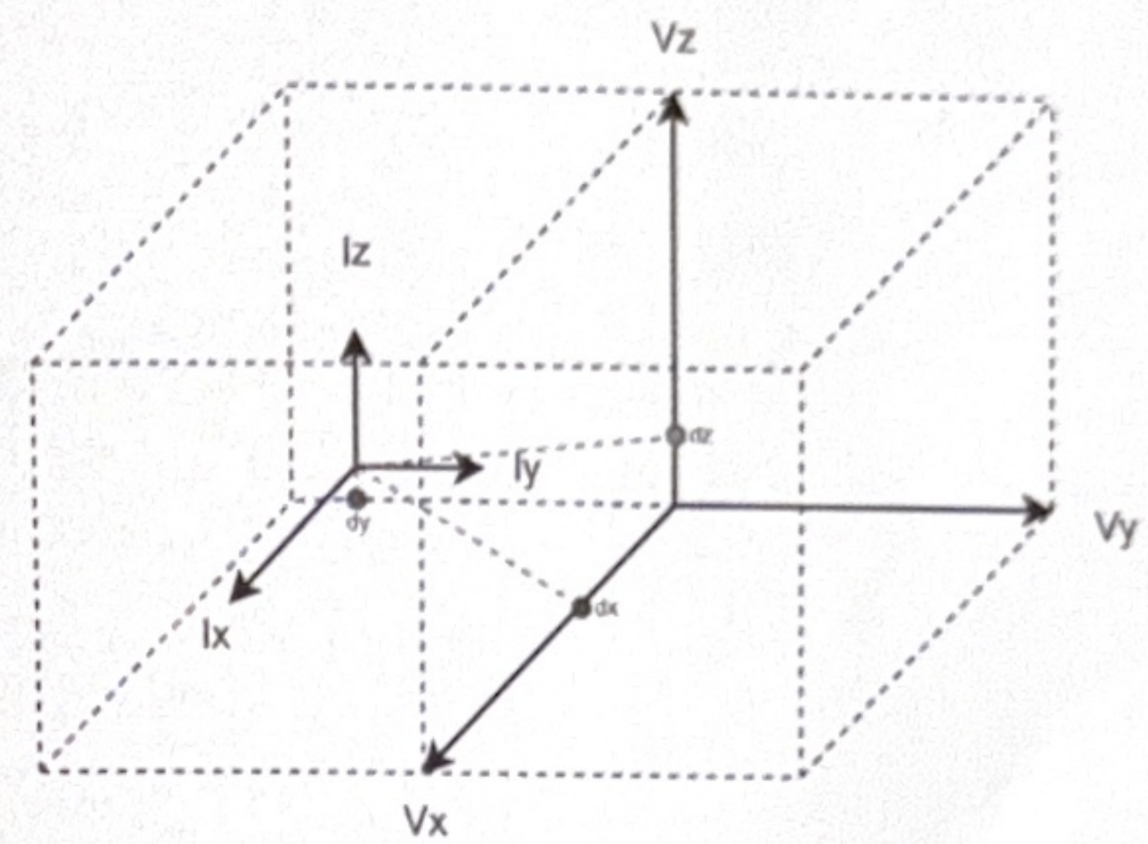
/3测试车板载IMU安装位置为例，以车辆质心为Rv圆点，imu在Rv系下坐标为(0.43,-0.38,0.2)

imu测量到的角速度与角速度转换至车辆质心

过之前的坐标系旋转，IMU坐标系已与车辆坐标系正交

为刚体上所有点的角速度都是相同的，不受到位置的影响，所以ωv=ωi

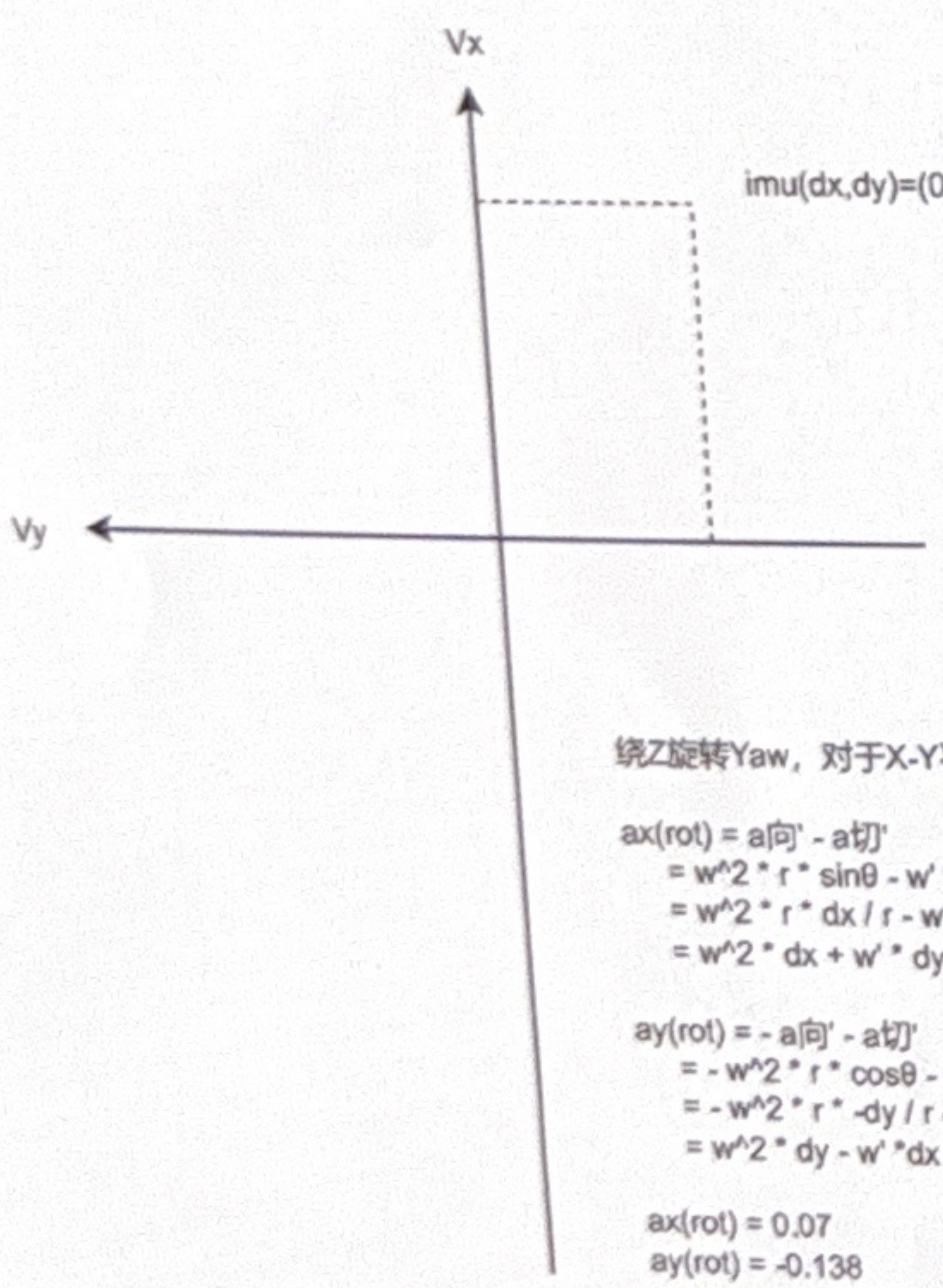
存在加速度的质心转换

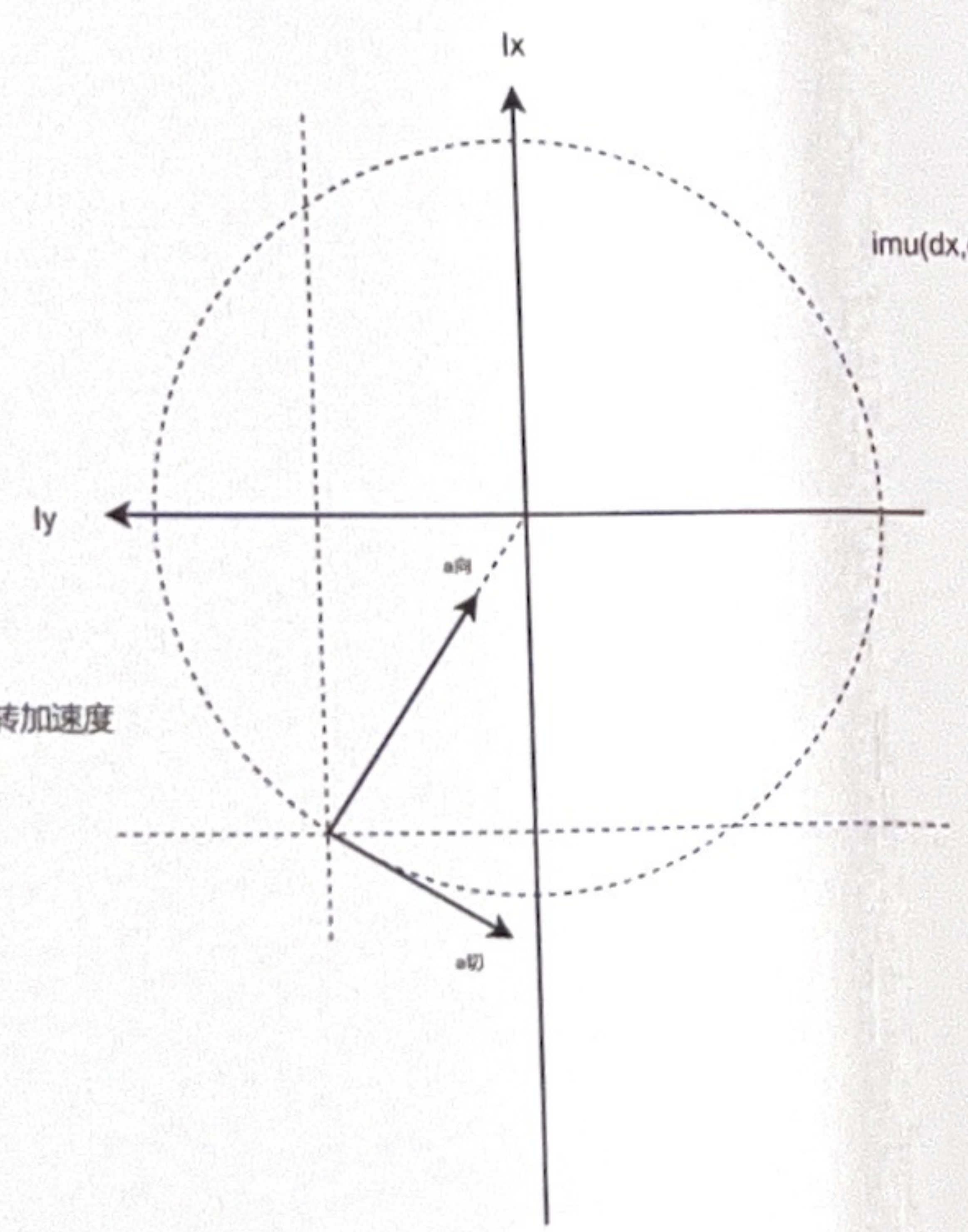


对于车辆运动，存在平动和旋转两种运动形式，那么imu测量到的加速度受到平动加速度和旋转加速度的影响
对于平动加速度：av(trans)=ai(trans)
对于旋转加速度：av(rot)=α **x** rv + ω **x** (ω **x** rv)，忽略科氏加速度，叉乘计算

计算加速度以"绕某一轴做平面旋转时，旋转加速度仅作用该平面"为原则进行手算

假设某时刻绕xyz轴角速度ω=(0.3,0.4,0.5)，角加速度α=(0.1,0.1,0.1)



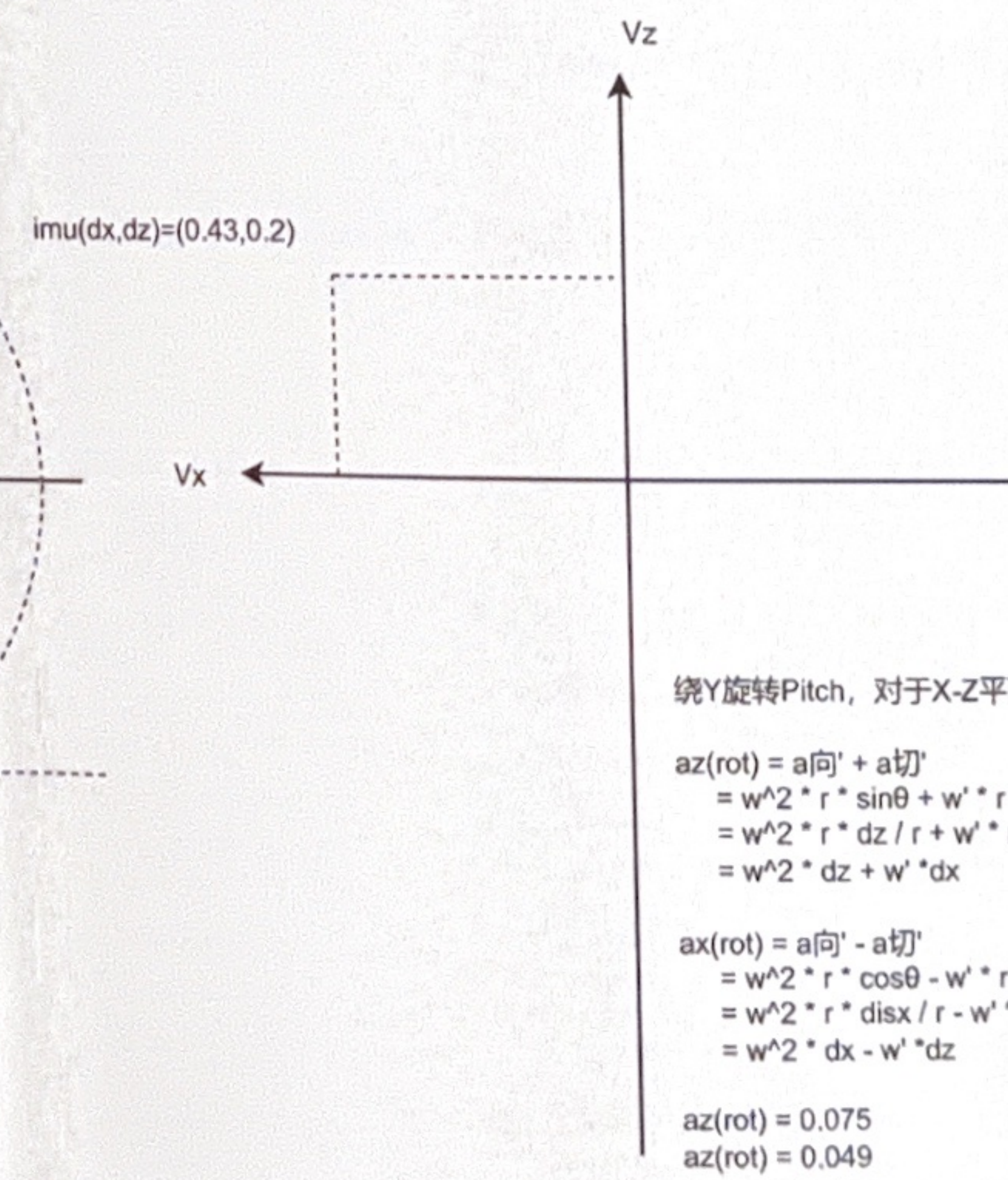


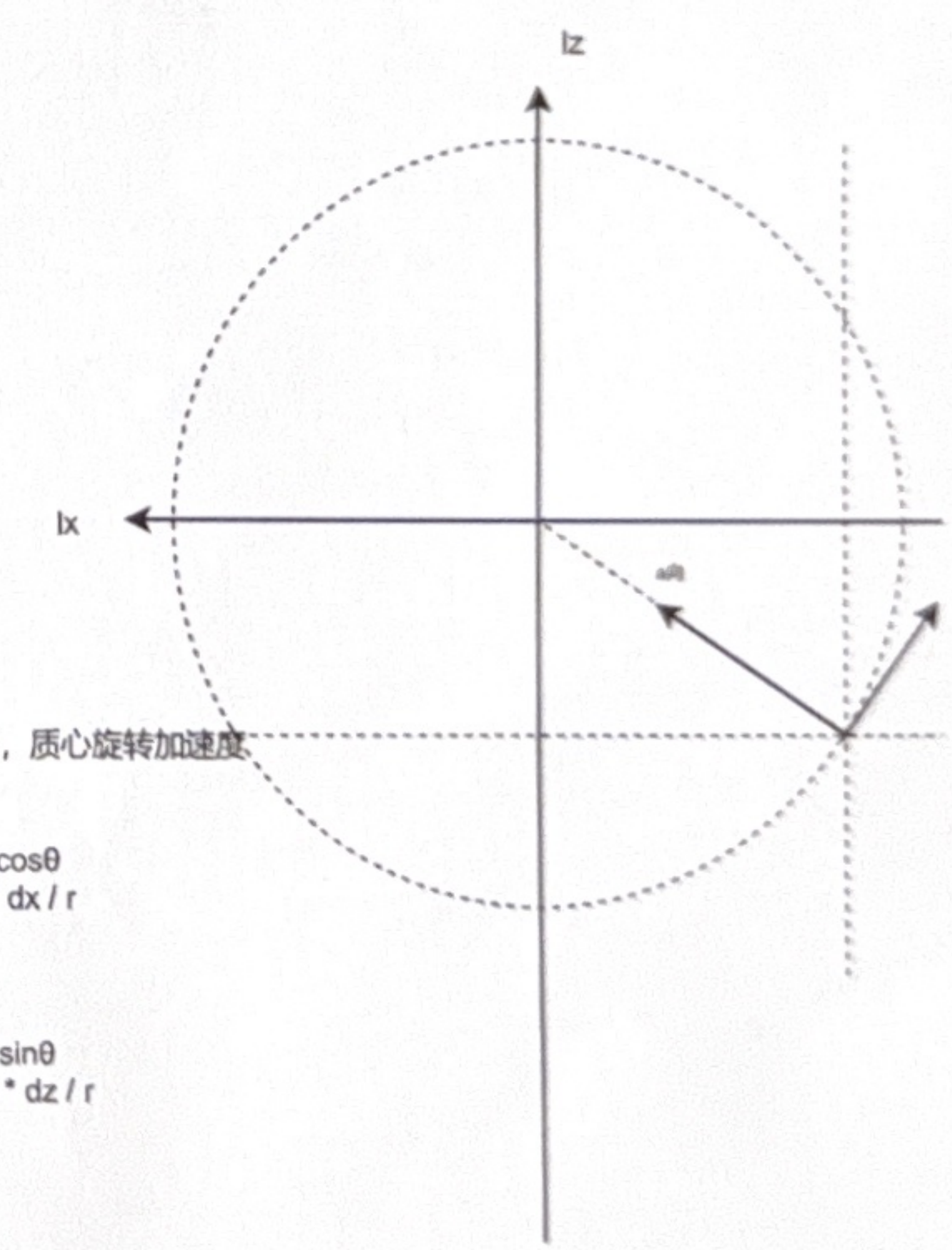
绕Z旋转Yaw，对于X-Y平面，质心旋转加速度

```
ax(rot) = a向' - a切'
    = w^2 * r * sinθ - w' * r * cosxθ
    = w^2 * r * dx / r - w' * r * -dy / r
    = w^2 * dx + w' * dy

ay(rot) = - a向' - a切'
    = - w^2 * r * cosθ - w' * r * sinθ
    = - w^2 * r * -dy / r - w' * r * dx / r
    = w^2 * dy - w' *dx

ax(rot) = 0.07
ay(rot) = -0.138
```



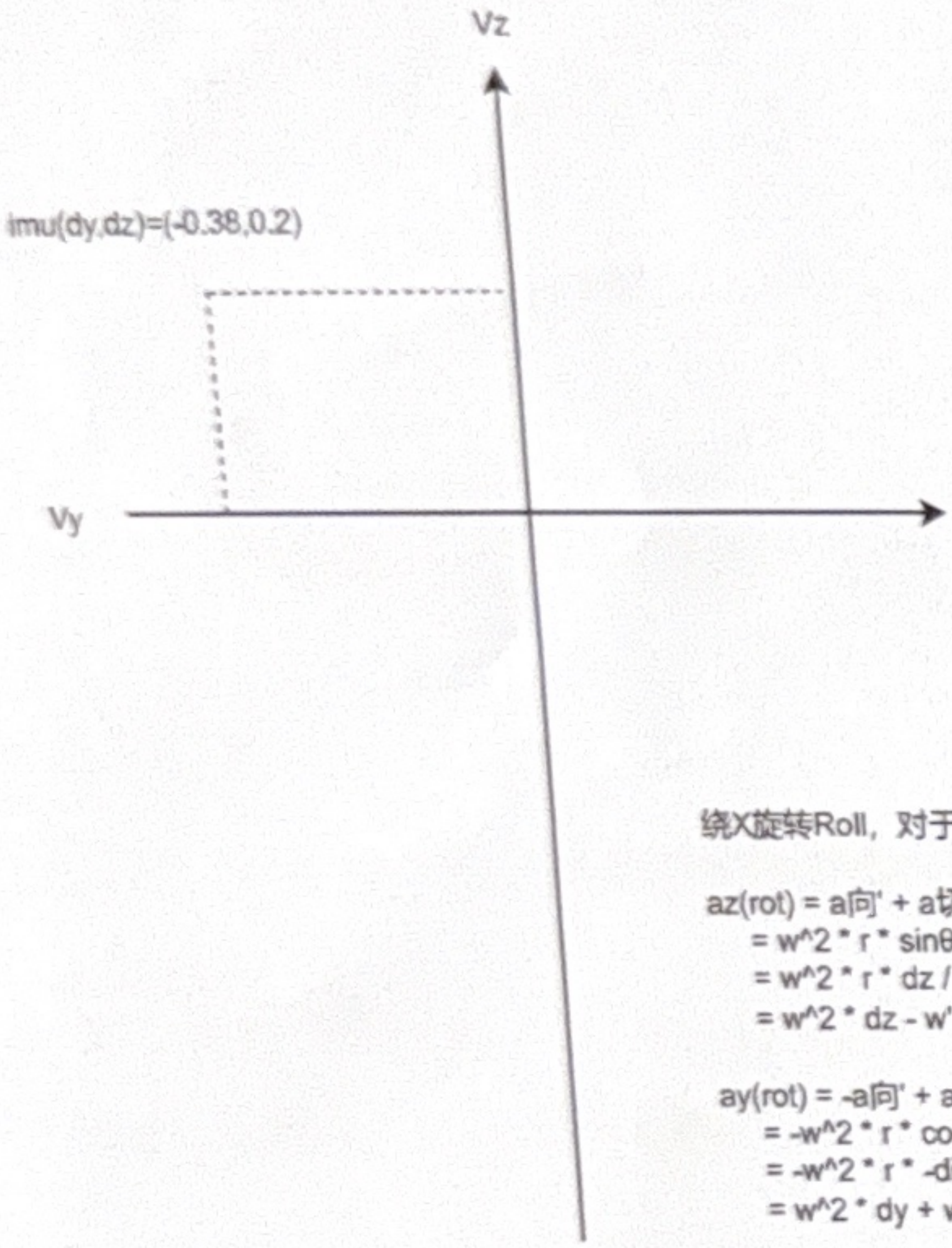


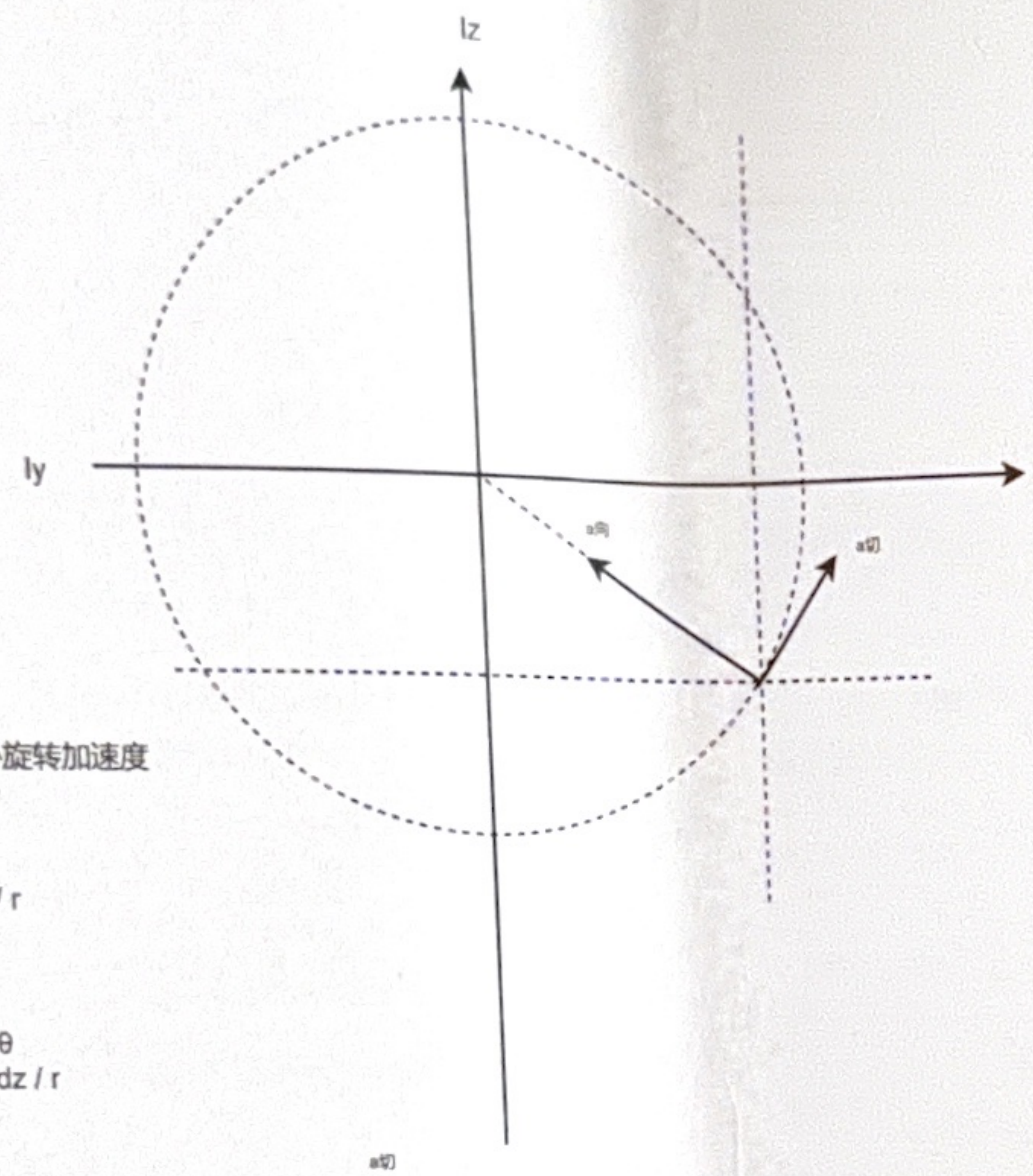
绕Y旋转Pitch，对于X-Z平面，质心旋转加速度

```
az(rot) = a向' + a切'
    = w^2 * r * sinθ + w' * r * cosθ
    = w^2 * r * dz / r + w' * r * dx / r
    = w^2 * dz + w' *dx

ax(rot) = a向' - a切'
    = w^2 * r * cosθ - w' * r * sinθ
    = w^2 * r * disx / r - w' * r * dz / r
    = w^2 * dx - w' *dz

az(rot) = 0.075
az(rot) = 0.049
```





绕X旋转Roll，对于Y-Z平面，质心旋转加速度

az(rot) = a向' + a切'
= w^2 * r * sinθ + w' * r * cosθ
= w^2 * r * dz / r + w' * r * -dy / r
= w^2 * dz - w' *dy

ay(rot) = -a向' + a切'
= -w^2 * r * cosθ + w' * r * sinθ
= -w^2 * r * -disy / r + w' * r * dz / r
= w^2 * dy + w' *dz

az(rot) = 0.056
ay(rot) = -0.014

质心相对于IMU的位置

$dx := -0.43$

$dy := 0.38$

$dz := -0.2$

$$\mathrm{Arotate}(dx, dy, dz, \omega x, \omega y, \omega z, d\omega x, d\omega y, d\omega z) := \begin{pmatrix} d\omega x \\ d\omega y \\ d\omega z \end{pmatrix} \times \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} + \begin{pmatrix} \omega x \\ \omega y \\ \omega z \end{pmatrix} \times \left[\begin{pmatrix} \omega x \\ \omega y \\ \omega z \end{pmatrix} \times \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} \right]$$

$$\mathrm{Arotate}(-2, 0, -1, 0, 5, 0, 0, 1, 0) = \begin{pmatrix} 49 \\ 0 \\ 27 \end{pmatrix}$$

$$\mathrm{Atrans}(aix, aiy, aiz) := \begin{pmatrix} aix \\ aiy \\ aiz \end{pmatrix}$$

+

$$Ao := \mathrm{Atrans} + \mathrm{Arotate}$$